

Software Developer Assessment: Maritime Operations & Compliance System

Context

You are building a platform for a marine organization to manage:

- Ship maintenance activities
- Safety drills and crew participation
- Compliance monitoring across ships

The system should help ensure that ships are **operationally safe and compliant with regulations**.

Objective

Build a **full-stack application** where:

- Admins manage maintenance tasks and safety drills
 - Crew members participate and log activities
 - The system tracks compliance and highlights risks
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Tech Stack (Flexible)

Frontend

- React (TypeScript preferred)

Backend

- Node.js / FastAPI / Spring Boot / Django

Database

- PostgreSQL / MongoDB
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Core Features

◆ 1. Ship Maintenance Module

Admin should be able to:

- Create maintenance tasks for ships
- Assign tasks to crew
- Update task status:
 - Pending
 - In Progress
 - Completed

Crew should be able to:

- View assigned tasks
 - Update status
 - Add notes/comments
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◆ 2. Safety Drill Management

Admin:

- Schedule safety drills (e.g., fire drill, evacuation)
- Assign drills to ships

Crew:

- View upcoming drills
 - Mark attendance
 - Submit drill completion
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◆ 3. Compliance Dashboard

System should:

- Show:
 - Pending maintenance tasks
 - Missed drills
 - Completed vs pending activities
 - Highlight:
 - Overdue maintenance
 - Missed safety drills
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Business Rules

- A maintenance task has a **due date**
 - A drill has a **scheduled date**
 - If not completed on time → marked as **non-compliant**
 - Compliance is calculated based on:
 - Completed maintenance %
 - Drill participation %
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Frontend Expectations

- Dashboard showing:
 - Maintenance summary
 - Drill summary
 - Compliance status
 - Pages:
 - Maintenance management
 - Drill management
 - Crew dashboard
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Bonus Features

- Role-based access control
 - Filters (by ship, status, date)
 - Notifications (overdue tasks)
 - Charts for compliance (basic graphs)
 - Docker setup
 - Deployment
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Evaluation Criteria

Technical

- API design
- DB schema
- Code structure

Logic

- Compliance calculation
- Handling overdue/missed cases

Frontend

- UI clarity
- Data visualization

System Thinking

- Scalability
 - Clean separation of concerns
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Submission Requirements

- GitHub repo
 - Business Flow of the application (DOCS,PDF)
 - README with:
 - Setup steps
 - Architecture decisions
 - Deployed link
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